

Prompt Engineering Syllabus

Institute	Archer Infotech Pune
Website	https://archerinfotech.in
Contact	+91 9850 678451 info@archerinfotech.in
Mode	Online and Offline batches at Kothrud, Pune
Duration	1 Month
Level	Beginner
Course page	https://archerinfotech.in/courses/generative-ai/prompt-engineering-training-in-pune

Course Overview

Learn practical prompt engineering for ChatGPT and AI tools, including role prompts, task design, evaluation and workflow templates. Duration: 1 Month. Level: Beginner. Includes live training, project practice, certificate, placement assistance, EMI options and a free demo class.

Research and Competitor Comparison

This advanced syllabus was revised after comparing Archer Infotech's website course catalogue with Pune competitor syllabi that emphasize project-based learning, full-stack/cloud coverage, testing, deployment, security, portfolio work, and AI-enabled workflows.

- Archer Infotech live course catalogue and Courses & Fees FAQ.
- Chedo Tech Java Full Stack syllabus: Java, Spring Boot, React, microservices, DevOps, cloud, AI integration.
- RI-TECH Java Full Stack syllabus: programming fundamentals, HTML/CSS, JavaScript, TypeScript, backend and AI.
- Ethan's DevOps Pune syllabus direction: CI/CD, cloud automation, hybrid delivery, next-gen AI-driven DevOps.
- SevenMentor Data Science and Full Stack Cloud pages: multi-disciplinary data, backend, cloud, dashboards, security, GenAI and ChatGPT integration.
- Infocampus, IT Education Centre, CACTS and Technogeeks Java Full Stack outlines: Java core, Spring Boot, Hibernate, React, APIs, Docker, AWS, testing and AI-enabled development.
- Pune competitor fee/course survey already stored in this vault for pricing and positioning context.

Detailed Topics

Topic 1: Competitor Benchmark and Archer Advantage

- Pune data competitors commonly combine Python, SQL, statistics, dashboards, machine learning, cloud, and GenAI tools.
- Advanced programs are moving toward deployable analytics systems, real dashboards, model evaluation, automation, and business storytelling.
- Archer positioning: teach data skills with practical business cases, explainable outputs, portfolio projects, and responsible AI usage.
- Outcome benchmark: student should convert raw data into validated insights, dashboards, models, or AI workflows that can be explained to business users.

Topic 2: Professional Toolchain and Workflow

- Industry-standard installation and environment setup.

- Git, GitHub, branching, pull requests, and code review workflow.
- IDE/editor productivity, debugging tools, extensions, and shortcuts.
- Documentation habits: README, setup guide, changelog, and learning log.
- Clean folder structure, naming conventions, and reusable templates.
- Comparison point: competitor syllabi increasingly bundle Git, projects, and portfolio work from the start.

Topic 3: Security, Quality, and Production Discipline

- Input validation and defensive programming or configuration checks.
- Common security risks for the subject and practical prevention habits.
- Testing strategy: unit, integration, manual, API, regression, or infrastructure checks as applicable.
- Error handling, logging, traceability, and root-cause thinking.
- Performance basics, resource usage, and user-experience impact.
- Production-readiness checklist before delivery or deployment.

Topic 4: AI and Generative AI Foundations

- AI, ML, GenAI, LLMs, agents, and copilots.
- Common business use cases.
- Capabilities and limitations.
- Responsible AI and privacy.
- Tool setup and safe usage.

Topic 5: Prompting Fundamentals

- Instruction clarity and context.
- Role, task, format, and examples.
- Prompt iteration and evaluation.
- Reusable prompt templates.
- Avoiding hallucinations and weak outputs.

Topic 6: LLM Workflows

- Research and summarization workflows.
- Coding and debugging assistance.
- Content and documentation generation.
- Data and spreadsheet assistance.
- Review and verification practices.

Topic 7: Automation and Tool Use

- AI-assisted workflows.
- Agents and multi-step tasks where relevant.
- Connecting tools and data safely.
- Human approval checkpoints.
- Audit trails and quality control.

Topic 8: Application and API Awareness

- LLM API concepts.
- Inputs, outputs, tokens, and cost.
- Basic retrieval and knowledge-base ideas.
- Evaluation and testing of AI outputs.
- Deployment awareness.

Topic 9: Business Implementation

- Use-case selection.
- Workflow mapping.
- Productivity measurement.
- Team adoption practices.
- Risk and compliance checklist.

Topic 10: Project Work

- Build or document a practical AI workflow.
- Create prompts, examples, and evaluation notes.
- Prepare a business-ready demo.
- Explain limitations and safeguards.
- Portfolio submission.

Topic 11: Advanced GenAI System Design

- Prompt patterns, context engineering, structured outputs, and evaluation rubrics.
- LLM APIs, token budgeting, latency, cost, safety, and model selection.
- RAG concepts: chunking, embeddings, vector search, citations, and answer verification.
- Agent workflows, tool use, human-in-the-loop approval, and audit trails.
- Guardrails, privacy, hallucination control, and responsible AI policies.

Topic 12: GenAI Business Automation Lab

- Create reusable AI workflows for content, coding, support, analysis, or operations.
- Build prompt libraries, checklists, and SOPs.
- Evaluate outputs against factual, style, and safety criteria.
- Measure productivity impact and failure cases.
- Prepare a deployable AI workflow demo.

Topic 13: Advanced Capstone for Prompt Engineering

- Build a realistic Prompt Engineering project or workflow aligned with Pune IT hiring expectations.
- Include requirements, architecture, implementation, tests, deployment or delivery notes, and troubleshooting evidence.
- Use GitHub with branches, commits, README, screenshots, setup instructions, and issue tracking.
- Add one production-style concern: security, performance, monitoring, automation, or user feedback.

- Present the capstone in interview format: problem, approach, stack, challenges, solution, and future scope.

Topic 14: Hands-on Practice and Assignments

- Topic-wise classroom and lab exercises.
- Small assignments after every major topic.
- Debugging, review, and correction sessions.
- Code or workflow explanation practice.
- Trainer feedback on practical submissions.

Topic 15: Mini Project Work

- One guided mini project based on the course technology.
- Requirement understanding, planning, implementation, and testing.
- Project folder organization and README writing.
- Final demo and viva-style explanation.
- Portfolio improvement suggestions.

Topic 16: Interview and Career Readiness

- Common interview questions for the subject.
- Scenario-based troubleshooting questions.
- Resume points and project explanation support.
- Mock interview and communication practice.
- Next learning path after course completion.

Topic 17: AI Integration for Prompt Engineering

- Use GenAI for dataset understanding, feature brainstorming, SQL/Python explanation, and dashboard narrative drafting.
- Build prompts that produce structured analysis plans, validation checklists, and stakeholder summaries.
- Use LLMs responsibly for model interpretation, synthetic test cases, and report review without leaking private data.
- Evaluate AI outputs for correctness, bias, hallucination, citation quality, and business usefulness.
- Final AI lab: create an AI-assisted analytics or GenAI workflow for Prompt Engineering.